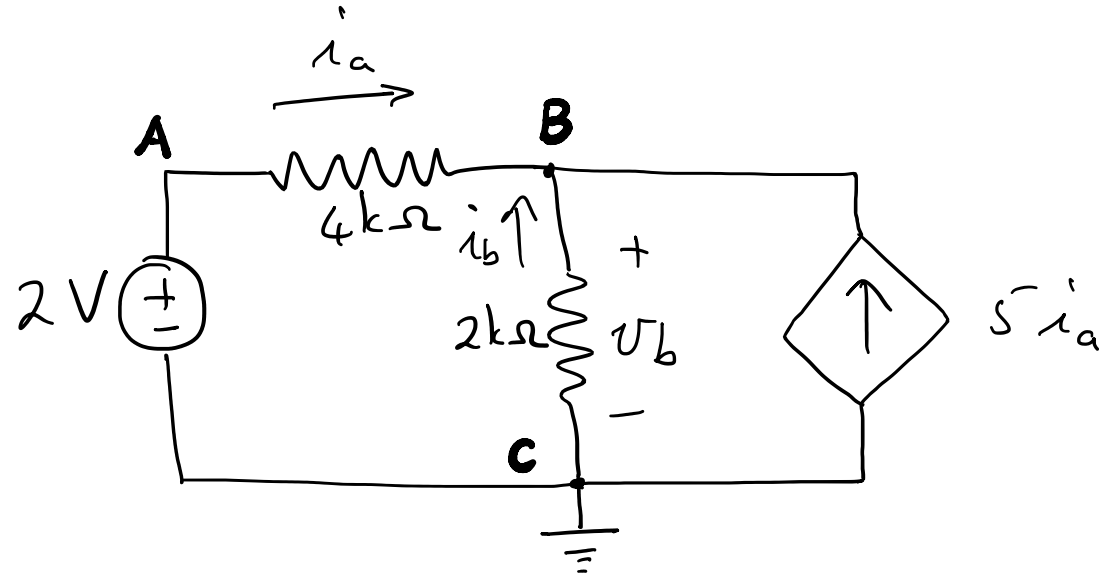


Question 1

Determine the voltage v_b



Answer:

$$\text{KCL: } i_a + 5i_a + i_b = 0$$

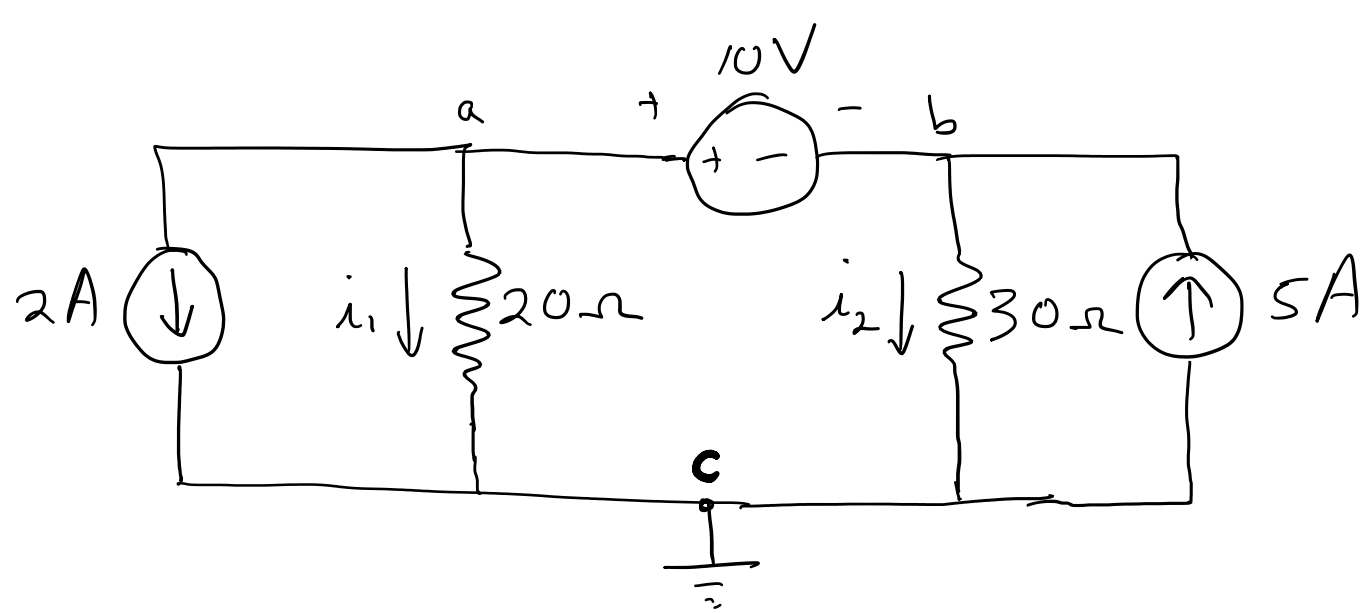
$$i_b = -\frac{v_b - v_c}{2k\Omega} = -\frac{v_b}{2k\Omega} = -\frac{v_b}{2k\Omega}$$

$$i_a = \frac{v_a - v_b}{4k\Omega} = \frac{2V - v_b}{4k\Omega}$$

$$v_b = \frac{3}{2}V$$

Question 2

Determine the node voltage v_a



Answer:

$$\text{KCL: } i_1 + i_2 + 2 = 5 \text{ (A)} \quad (1)$$

$$i_1 = \frac{v_a}{20\Omega} \quad (2)$$

$$i_2 = \frac{v_b}{30\Omega} \quad (3)$$

$$v_a - v_b = 10V \quad (4)$$

$$\Rightarrow v_b = v_a - 10V \quad (5)$$

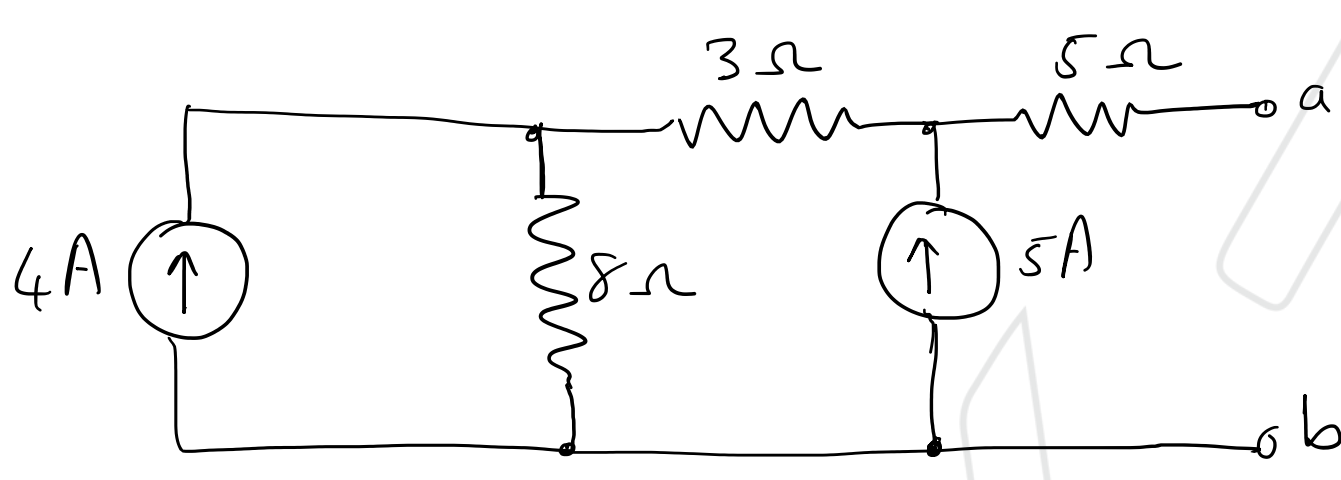
$$\Rightarrow i_2 = \frac{v_a - 10V}{30\Omega} \quad (6)$$

Substitute (6) and (2) into (1) and solve for v_a :

$$\Rightarrow v_a = 40V$$

Question 3

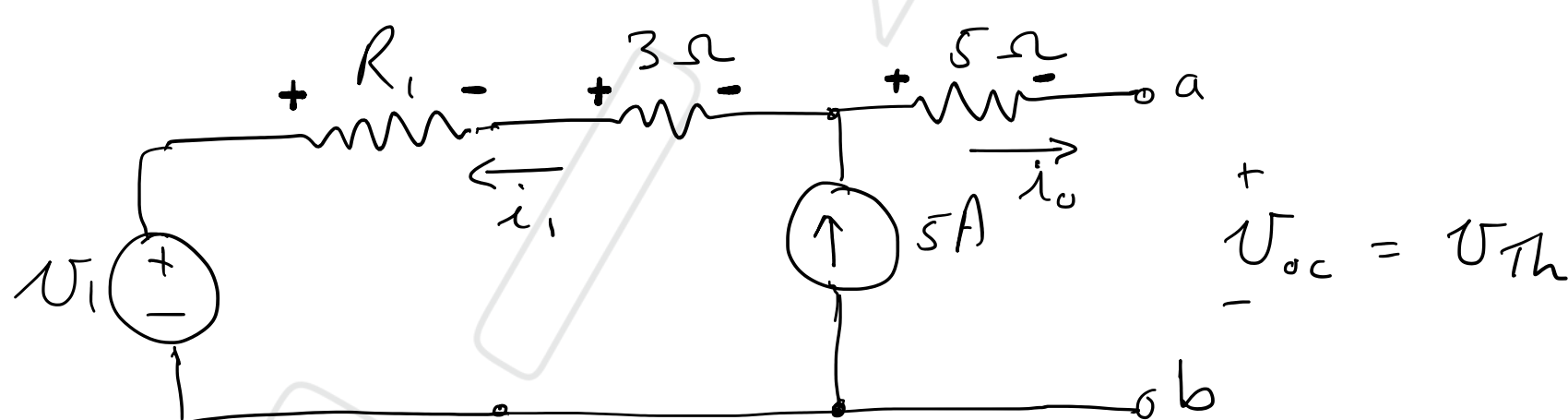
Consider the circuit:



(a) Compute the Thevenin voltage.

Answer

The circuit above is equivalent to



$$R_1 = 8\Omega$$

$$V_1 = 4A \cdot 8\Omega = 32V$$

$$i_o = 0A$$

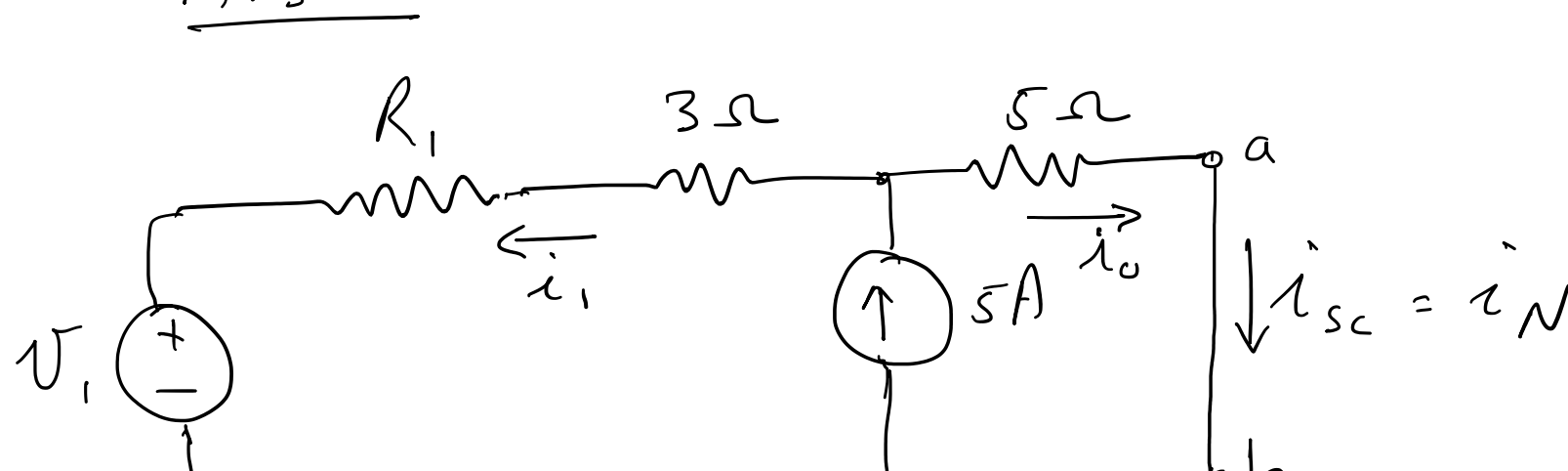
$$i_1 = 5A$$

$$\text{KVL: } V_{Th} = i_o \cdot 5\Omega + i_1 \cdot (3\Omega + R_1) + V_1$$

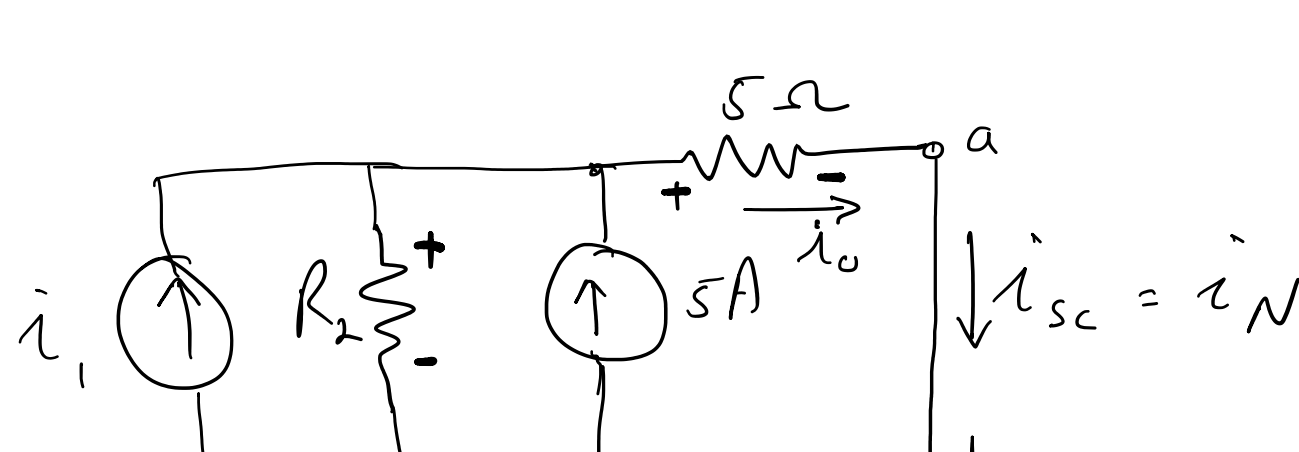
$$= 87V$$

(b) Compute the Norton Current

Answer



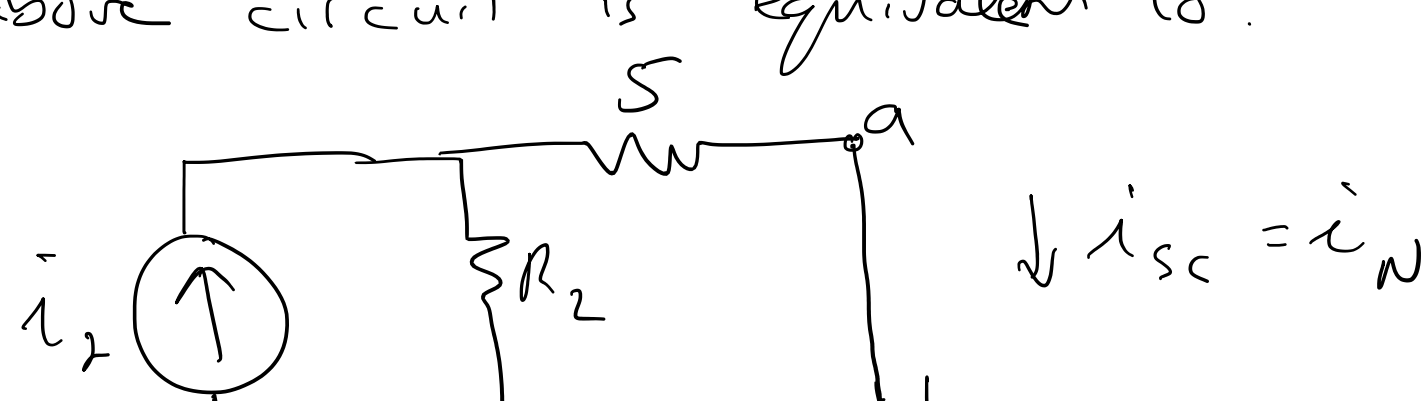
The above circuit is equivalent to



$$R_2 = R_1 + 3\Omega = 11\Omega$$

$$i_1 = \frac{V_1}{R_1 + 3\Omega} = \frac{32}{11}A$$

The above circuit is equivalent to:



$$i_2 = i_1 + 5A = \frac{87}{11}A$$

The above circuit is equivalent to:



$$V_2 = R_2 i_2 = 87V$$

$$i_N = \frac{V_2}{R_2 + 5\Omega} = \frac{87}{16}A$$

(c) Compute the Thevenin resistance.

$$R_{Th} = \frac{V_{Th}}{i_N} = 16\Omega$$

(d) Compute the load resistance to maximise power absorbed by the load.

$$R_L = R_{Th} = 16\Omega$$